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Giardia duodenalis induces pathogenic dysbiosis of human intestinal microbiota biofilms

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ABSTRACT

Giardia duodenalis is a prevalent cause of acute diarrheal disease worldwide. However, recent outbreaks in Italy and Norway have revealed a link between giardiasis and the subsequent development of chronic post-infectious irritable bowel syndrome. While the mechanisms underlying the causation of post-infectious irritable bowel syndrome remain obscure, recent findings suggest that alterations in gut microbiota communities are linked to the pathophysiology of irritable bowel syndrome. In the present study, we use a laboratory biofilm system to culture and enrich mucosal microbiota from human intestinal biopsies. Subsequently, we show that co-culture with *Giardia* induces disturbances in biofilm species composition and biofilm structure resulting in microbiota communities that are intrinsically dysbiotic – even after the clearance of *Giardia*. These microbiota abnormalities were mediated in part by secretory-excretory *Giardia* cysteine proteases. Using in vitro cell culture and germ-free murine infection models, we show that *Giardia*-induced disruptions of microbiota promote bacterial invasion, resulting in epithelial apoptosis, tight junctional disruption, and bacterial translocation across an intestinal epithelial barrier. Additionally, these dysbiotic microbiota communities resulted in increased activation of the Toll-like receptor 4 signalling pathway, and overproduction of the pro-inflammatory cytokine IL-1beta in humanized germ-free mice. Previous studies that have sought explanations and risk factors for the development of post-infectious irritable bowel syndrome have focused on features of enteropathogens and attributes of the infected host. We propose that polymicrobial interactions involving *Giardia* and gut microbiota may cause persistent dysbiosis, offering a new interpretation of the reasons why those afflicted with giardiasis are predisposed to gastrointestinal disorders post-infection.
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1. Introduction

Giardia duodenalis is a leading cause of diarrheal disease worldwide, and was recently included in the World Health Organization (WHO) Neglected Disease initiative (Buret, 2007; Hanevik et al., 2009). Moreover, *Giardia* is emerging as a prominent precursor to post-infectious irritable bowel syndrome (PI-IBS) and a variety of chronic extra-intestinal disturbances such as chronic fatigue (Hanevik et al., 2009; Wensaas et al., 2012; Halliez and Buret, 2013). Acute bacterial gastroenteritis, as well as giardiasis, is now a well-established risk factor in the development of PI-IBS (Spiller and Campbell, 2006; Thabane and Marshall, 2009; Marshall et al., 2010; Cremon et al., 2014). Much remains to be elucidated about the multifactorial pathophysiological consequences following acute enteric infection. Recent findings have associated dysbiosis of faecal microbiota with a number of complications (Salonen et al., 2010; Carroll et al., 2011; Rajilic-Stojanovic et al., 2011). Thus, extending beyond the basic principles of Koch's postulate and association studies, research now needs to determine whether and how pathogen-induced dysbiosis of mucosal microbiota biofilms may cause disease long after the inciting microorganism has been cleared.

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We hypothesised that enteropathogens may be able to alter commensal microbiota biofilm bacteria during the acute phase of infection, possibly promoting their ability to act as opportunistic pathogens, which in turn would be able to trigger disease in the absence of the initiating pathogen. Indeed, *Campylobacter jejuni*, a well-known cause of PI-IBS as well as inflammatory flares in patients with inflammatory bowel disease (IBD), induces the translocation of otherwise non-invasive commensal bacteria, both via paracellular and transcellular routes, including a novel pathway hijacking host lipid rafts (O'Hara and Buret, 2008; Kalischuk et al., 2009; Abreu, 2010; Kalischuk et al., 2010). It was recently found that exposure to *C. jejuni* secretory-excretory products was able to activate latent virulence genes in non-invasive *Escherichia coli* (Reti et al., 2015). Recent observations indicate that during the acute phase of giardiasis, microbiota may contribute to CD8 T lymphocyte-mediated impairment (Keselman et al., 2016). However, whether and how *Giardia* may directly modify gut microbiota, and whether disrupted microbiota on their own may cause intestinal abnormalities, remains to be elucidated (Keselman et al., 2016). In an attempt to further characterise the mechanisms involved in this novel concept of microbially induced chronic pathogenesis, the present study investigated how *Giardia* may alter host mucosal microbiota biofilms to cause post-infectious gut abnormalities in the absence of the parasite.

Within the healthy human gastrointestinal tract the mucosa exists in close association with multispecies biofilms encompassing the commensal microbiota, but much remains unknown on how the integrity and physiology of these communities affect gut homoeostasis and disease (Kleessen and Blaut, 2005; von Rosenvinge et al., 2013). These mucosal biofilm bacteria are different from those living in the intestinal lumen (Macfarlane and Macfarlane, 1997). Indeed, intestinal biofilm bacteria growing on intestinal mucin differ metabolically and phylogenetically from those living in a planktonic state (Macfarlane et al., 2005). Studies found that genetic exchange between representatives of the gut microbiota are common, which raises the question as to whether or not such exchanges may also occur between enteropathogens and the commensal microbiota (Zoetendal et al., 2008). Intrinsic to the ability of bacteria to persist in biofilms is the coordinated secretion by constituent cells of substances constituting an extracellular matrix (ECM), of which polysaccharides, proteins and DNA are the largest contributors (Sutherland, 2001; Dongari-Bagtzoglou, 2008). Recent studies have found that planktonic bacteria released from biofilms may exhibit increased virulence (Chua et al., 2014). We hypothesised that exposure of the epithelium to abnormal release and proliferation of planktonic bacterial species, which in turn may act as opportunistic pathobionts, may prove detrimental to homoeostatic host-microbiota symbiosis.

Here we use an in vitro technique to culture and enrich microaerobic biofilm communities directly from human intestinal mucosal biopsy specimens (Sproule-Willoughby et al., 2010). This study exploits *G. duodenalis* as a model organism to investigate the impact of an acute human enteropathogen on the structure and community composition of these ex vivo human mucosal microbiota biofilms. We specifically aimed to: (i) assess the community composition and structure of human microbiota biofilms following exposure to *Giardia*, (ii) investigate the impact of *G. duodenalis*-perturbed microbiota biofilms, in vivo, using humanized germ-free mice, and (iii) determine whether *Giardia* interactions with human microbiota biofilms may induce changes in epithelial monolayer integrity and function. The results indicate that *Giardia* causes mucosal microbiota dysbiosis, and that these dysbiotic microbiota on their own may disrupt human enterocytic epithelia, and cause intestinal abnormalities in humanized mice. Based on these findings, we postulate that *Giardia*-induced post-infectious microbiota dysbiosis may lead to intestinal disease.

2. Materials and methods

2.1. Growth medium, reagents and antibodies

Tryptic soy broth (BD Biosciences, USA) served as the base for medium used to grow microbiota biofilm communities, and was supplemented with 4 g/L of yeast extract (BD Biosciences), 1 mL/L of 5% L-cysteine-HCl (Sigma, USA) solution, and 1 mL/L of hemin/ menadione solution (1 mg/mL menadione in 95% Ethanol (Sigma) and 50 mg of bovine hemin (Sigma), 1.74 g of K_2HPO_4, 0.4 g of NaOH, all dissolved in distiled deionised H_2O)). *Giardia* trophozoites were cultivated in Diamond's TY1-S-33 medium (sTSY) in 15 mL polystyrene centrifuge tubes. Caco-2 intestinal epithelial cells were grown in Minimal Essential Medium Eagle (MEME; Sigma), supplemented with 20% heat-inactivated FBS, 10 ml/L of penicillin–streptomycin stabilized solution, 2 mM L-glutamine and 1 mM sodium pyruvate (all from Sigma). For some experiments, Caco-2 monolayers were incubated with purified IL-1β (5 ng/mL) (R&D Systems, Minneapolis, USA) for 2 h.

The following inhibitors were used for different experiments: broad spectrum cysteine protease inhibitor E-64d (1 μM) (Sigma), and Cathepsin B inhibitor Ca-074Me (1 μM) (Peptides International, Louisville, Kentucky, USA). Inhibitors were used to pretreat *Giardia* 3 h prior to preparation of conditioned medium, and remained in the medium throughout the infection/treatment period.

The following antibodies were used for immunocytochemistry, immunohistochemistry, or immunoblotting: mouse monoclonal anti-glyceraldehyde 3 phosphate dehydrogenase (GAPDH; 1:1000) (Santa Cruz Biotechnology, Santa Cruz, USA), rabbit monoclonal anti-zonula occludens (ZO)-1 (1:500/1:10,000), mouse monoclonal anti-Toll-like receptor (TLR)4 (1:500/1:10,000), rat polyclonal anti-CD45R (1:10,000), rat monoclonal antibody against Thy-1.2 (CD90), all from Abcam Inc. (Cambridge, USA). Mouse Alexa Fluor 555 and Alexa Fluor 488-conjugated primary antibodies (Invitrogen, USA) were used for immunostaining (1:2000); negative controls were IgG from rat coupled with the same fluorochrome. Mouse and rabbit (Cell Signaling, USA) horseradish peroxidase (HRP)-conjugated secondary antibodies were used for western blotting (1:1000).

2.2. Colonic biopsy collection and processing

Ongoing research in our laboratory indicates that giardiasis affects the host's large and small intestine, making it very relevant to investigate effects of *Giardia* on the microbiota along the entire intestine (Amat et al., 2015; Halliez et al., 2016). Also, as gut microbiota are much more abundant in the large versus small intestine, we examined the impact of *Giardia* on colonic microbiota communities. Mucosal biopsies from the descending colon of healthy human volunteers were collected during routine colon cancer screening procedures. Biopsy samples were obtained through the Intestinal Inflammation Tissue Bank at the University of Calgary, Canada. The specific protocols used for this study underwent ethics approval with the University of Calgary review board. With patients' informed consent, biopsies were collected into BBL Part-A-Cul tubes (BD Biosciences) in order to maintain an anaerobic environment, and processed according to the manufacturer's instructions (Sproule-Willoughby et al., 2010). In an anaerobic chamber (90% nitrogen, 5% hydrogen, 5% carbon dioxide gases), biopsies were washed in ethanol, 0.016% DTT, and PBS. Next, biopsies were homogenised in 200 μl of supplemented Diamond's TY1-S-33 medium (sTSY) (Pellet Pestle® Microgrinder System; Kimble-Kontes), and then 500 μl of 50% glycerol were added and samples were aliquoted into six tubes and stored at −70 °C until the seeding of biofilms onto the Calgary Biofilm Device (CBD,

commercially available as the MBEC™ Assay, Innovotech Inc., Canada).

2.3. Microbiota biofilm formation(MBBF) from colon biopsy samples

Enrichment of microbiota from intestinal biopsies was accomplished by established methods (Sproule-Willoughby et al., 2010) using the CBD (Ceri et al., 1999; Harrison et al., 2010). This ex vivo culture technique enriches representative bacteria from most bacterial classes known to reside in the human gastrointestinal tract, and the resulting biofilms are stable and available for study after times exceeding 6 days of incubation (Sproule-Willoughby et al., 2010). Briefly, processed biopsy samples stored at −70 °C were transferred into 9.8 mL of sTSY, which contained tryptic soy broth (BD Biosciences Inc.) supplemented with 4 g/L of yeast extract, 1.74 g/L of K_2HPO_4, 0.4 g/L of NaOH and 1 mL/L of 5% L-cysteine-HCl, and 1 mL/L of hemin-menadione solution (containing 50 mg/mL of bovine hemin and 1 mg/mL of menadione in 95% ethanol). This inoculum was further diluted 1:10 with sTSY and 150 μL aliquots added to each of the 96 wells of the CBD for biofilm formation. Inoculated CBDs were sealed inside anaerobic bags (AnaeroGen™ Compact System, Oxoid, Napean, ON, Canada), placed on an orbital shaker at 75 rpm (Model G2; New Brunswick Scientific Company, Canada), and incubated at 37 °C for 72 h.

2.4. Parasites

Giardia duodenalis strain NF, obtained from an outbreak of human giardiasis in Newfoundland, Canada (Teoh et al., 2000), was used for all experiments. Trophozoites were grown at 37 °C in Diamond TY1-S-33 medium in 15 mL polystyrene centrifuge tubes, and sub-cultured every 2–3 days in order to maintain the lines. Trophozoites were harvested at log phase after cold shock on ice for 30 min, followed by centrifugation at 300*g* for 10 min at 4 °C. Medium was subsequently removed, together with dead parasite sediments, via aspiration, and the remaining pellet washed once with sterile PBS. The final pellet was resuspended in 1 mL of sTSY, and the trophozoite concentration was determined by counting using a haemocytometer and microscopy, followed by an adjustment to 1×10^7 trophozoites/mL. Experiments were performed using both live *Giardia* and spent sTSY medium, in order to determine parasite impact on biofilm communities in the presence/absence of physical interference. Trophozoite conditioned medium was prepared by inoculating sTSY with trophozoites (1×10^7 trophozoites/mL), and incubation at 37 °C for 24 h at 75 rpm on an orbital shaker. Trophozoites were subsequently removed via centrifugation. For the preparation of live trophozoite challenge, trophozoites were harvested as described above, and added to challenge plates (1×10^7 trophozoites/mL in sTSY).

2.5. Microbiota biofilm-Giardia challenge

Giardia trophozoites and spent medium were prepared as described in Sections 2.1 and 2.4. Live trophozoite preparations, spent medium, or sterile sTSY were added to 96-well microtiter "challenge" plates (Corning, USA) at 150 μl/well, to which CBD lids containing formed MBBF were subsequently transferred. The challenge plates were incubated for an additional 24 h at 37 °C, after which pegs were removed from the CBD lids, and planktonic bacteria were collected for viability and numerical colony forming units (CFU) counts (Supplementary Fig. S1), structural integrity (confocal scanning laser microscopy and scanning electron microscopy), extracellular matrix integrity (scanning electron microscopy and Wheat-germ Agglutinin (WGA) assay), and species profiling (16S rRNA Illumina (Canada) sequencing) (Harrison et al., 2006).

2.6. In vitro cell culture model

All in vitro experiments were done using the human epithelial colorectal adenocarcinoma cell line, Caco-2 (American Type Culture Collection, www.atcc.org). The Caco-2 cell line represents a widely used model for intestinal barrier function, and forms a monolayer of cells which mimic mature enterocytes, and expresses many of their functional characteristics and cellular markers (Sambuy et al., 2005). Cells were kept at 37 °C with 5% CO_2 in 96% humidity, as previously described (Lapointe and Buret, 2012). Culture medium was replenished every 2–3 days, and 2× washes of trypsin-EDTA (Sigma) treatment was used to passage every 6–8 days. Six-well culture-treated plates (Becton Dickinson, Canada), Lab-Tek chamber slides (Nalge Nunc International, USA), or Transwell filter units with semipermeable filter membranes (0.4- to 8.0-μm pores; Costar, USA) were seeded with trypsinized cells (5×10^4 cells per ml). In all studies, Caco-2 cells were used between passages 23 and 27 from the stock aliquots in the laboratory. At the time of treatment, Caco-2 cell monolayers were incubated in reduced serum medium (Opti-MEM®).

2.7. In vitro biofilm and epithelial cell co-culture

For assessment of apoptosis (ELISA) and characterisation of TLR4 expression and tight junctional ZO-1 integrity, epithelial monolayers were used 9 days after passage, when they were confluent but not overgrown. Permeability was measured in Transwell filter units on monolayers after 9 days when electrical resistance reached >500 $\Omega \cdot cm^2$. At the time of the experiment, monolayers were incubated in Opti-MEM (Gibco, USA). To assess the effects of *Giardia*-induced microbiota alterations on enterocytes, and to determine whether planktonic-swimming bacteria released by these disrupted microbiota biofilms were in fact pathogenic, planktonic constituents of MBBF were collected, following challenge, via centrifugation, as previously described (Harrison et al., 2006; Sproule-Willoughby et al, 2010). Bacterial pellets were resuspended in 300 μl of Opti-MEM, and 100 μl of the solution were added to each confluent Caco-2 monolayer and allowed to incubate at 37 °C with 5% CO_2 in 96% humidity for various time points (1–24 h).

2.8. Scanning electron microscopy (SEM)

Preparations were examined by SEM to characterise biofilm ECM surface abnormalities induced by *G. duodenalis*. Pegs from the CBD lid were carefully broken off with sterile pliers, and fixed in 2.5% glutaraldehyde in 0.1 M cacodylate buffer (Sigma) for 3 h. Pegs were then air dried for 4–5 days, and subsequently mounted on aluminium stubs and sputter coated with gold–palladium as previously described (Harrison et al., 2006). Photomicrographs were obtained with a FEI XL30 scanning electron microscope at an acceleration voltage of 30 kV.

2.9. WGA assay

Fluorescently-labelled WGA (Molecular Probes, USA) was used to detect N-acetylglucosamine residues, known markers of the ECM in microbiota biofilm communities (Perez-Mendoza et al., 2011). CBD lids containing MBBF communities were transferred into a 96-well plate containing WGA stain, and left at 4 °C for 2 h. CBD lids were transferred into a new 96-well plate containing 200 μl of 33% acetic acid and subjected to sonication (on high setting) for 10 min in an Aquasonic sonicator. A low output sonicator was used to allow for dispersal of biofilms into the acetic acid without affecting cell viability (Ceri et al., 1999). Plates were then

incubated at 37 °C for 1 h, followed by an additional sonication stage and fluorometric analysis of the resulting ECM components sloughed off into the acetic acid using a Spectramax microplate reader (Molecular Devices Corp., Menlo Park, USA).

2.10. XTT assays

Bacterial viability was determined via reduction of the tetrazolium salt XTT (Sigma). XTT has previously been shown as a reliable indicator of bacterial viability (Roslev and King, 1993). A 5 mM stock solution was prepared, and stored at 4 °C until use, at which time it was diluted 1:100. The resulting XTT solution was supplemented with Menadione (10 mM). Calgary biofilm device (CBD) pegs containing biofilms were transferred to 150 µl of XTT solution and incubated at 37 °C for 2 h. Planktonic constituents were pooled (three wells) and centrifuged at 8000*g* (10 min). The supernatant was discarded, and the resulting pellet resuspended in 200 µl of XTT/menadione solution, and incubated at 37 °C for 1 h. Following incubation, the XTT solution was collected into 96-well plates and absorbance (450 nm) was determined using a SpectraMax microplate reader.

2.11. Cysteine protease assays

Cathepsin cysteine protease activity was determined using the fluorogenic cysteine protease substrate Z-Phe-Arg-AMC (200 µM) (Peptides International, Louisville, KY, USA) in characterisation experiments.

The following inhibitors were used for different experiments: broad-spectrum cysteine protease inhibitor E-64d (1 µM) (Sigma), and Cathepsin B inhibitor Ca-074Me (1 µM) (Peptides International). Inhibitors were used to pre-treat *Giardia* 3 h prior to preparation of conditioned medium, and remained in the medium throughout the infection/treatment period.

Cathepsin cysteine protease activity in supernatants of *Giardia*-spent medium, and in biofilm/*Giardia*-conditioned medium co-cultures, was measured via liberation of 7-aminomethylcoumarin (AMC) from fluorogenic substrates, using a Spectramax microplate reader (Molecular Devices Corp.). This method has been validated through cysteine protease-mediated proteolytic processing, and the change in reflective light units (RFUs), which can be used to calculate slope (Barrett, 1980; Tchoupe et al., 1991). Assays were performed at pH 7.2 in Cathepsin buffer (100 mM sodium acetate, 10 mM DTT, 0.1% Triton X-100, 1 mM EDTA, 0.5% DMSO – all from Sigma) in the presence/absence of broad-spectrum cysteine protease inhibitor E-64, and Cathepsin B inhibitor Ca-074Me, at concentrations of 1 µM, which was previously shown to have no detrimental effect on trophozoite viability (Rodriguez-Fuentes et al., 2006).

2.12. Apoptosis

Enterocyte apoptosis was quantified in Caco-2 monolayers, following co-incubation with planktonic MBBF (+/− *Giardia* spent medium) using a Cell Death Detection ELISA Kit (Roche Diagnostics, Laval, Canada). During apoptosis, mono- and oligonucleosomes are released, and this assay detects the nucleosome-specific histone regions (H1, H2A, H2B, H3 and H4). Measurements were taken in triplicate (405 nm) from 10^5 enterocytes per group, and plates were read at 5 min intervals using a SpectraMax microplate reader. Values were expressed as absorbance ratios of the experimental lysates versus absorbance calculated for CaCo-2 cells exposed to control MBBF (arbitrarily set at 1.0) The ELISA has a detection limit of 10^2 apoptotic cells.

2.13. In vitro permeability

Translocation of planktonic MBBF constituents was measured in Transwell filter units on monolayers after 9 days when electrical resistance reached >500 $\Omega \cdot cm^2$ in control monolayers. Planktonic constituents of MBBF were collected from the CBD, following challenge with *Giardia* spent medium or vehicle (medium containing no *Giradia*) via centrifugation. Bacterial pellets were resuspended in 300 µl of Opti-MEM, and 100 µl of the solution was added to confluent Caco-2 monolayers and allowed to incubate at 37 °C with 5% CO_2 in 96% humidity for 24 h. Two hundred microlitre were collected from the basolateral compartment, and serially diluted onto spot-plates (Luria-Bertani (LB) agar) for colony-forming enumeration following incubation at 37 °C for 24 h.

2.14. ZO-1, TLR-4

The following antibodies were used for immunocytochemistry, immunohistochemistry, or immunoblotting: mouse monoclonal anti-GAPDH (1:1000) (Santa Cruz Biotechnology), rabbit monoclonal anti-ZO-1 (1:500/1:10,000), mouse monoclonal anti-TLR4 (1:500/1:10,000), rat polyclonal anti-CD45R (1:10,000), rat monoclonal antibody against Thy-1.2 (CD90), all from Abcam Inc.. Mouse Alexa Fluor 555 and Alexa Fluor 488-conjugated primary antibodies (Invitrogen) were used for immunostaining (1:2000), negative controls were IgG from rat coupled with the same fluorochrome. Mouse and rabbit (Cell Signaling) HRP-conjugated secondary antibodies were used for western blotting (1:1000). Caco-2 cell monolayers, grown to confluence on 6-well plates, were co-cultured (as described above) with MBBF planktonic constituents after exposure to *Giardia* spent medium. Following co-culture (2 h) cells were lysed, and protein normalised to 2 mg/mL (Bradford assay; Bio-Rad Laboratories, Hercules, USA), diluted 1:1 in electrophoresis buffer and boiled (3 min). Samples were then separated by SDS-PAGE (10%) and transferred to nitrocellulose membranes (Whatman, UK). Membranes were blocked (5% non-fat dry milk powder in Tris-buffered saline (TBS)+0.1% Tween (T)) for 1 h, followed by incubation with primary antibodies for ZO-1, Occludin and TLR-4 (all from Abcam Inc.) in 5% milk TBS-T overnight at 4 °C. Membranes were subsequently washed (three times) with TBS-T and then conjugated with the appropriate secondary antibody for 1 h. Bands were visualised using ECL-plus chemiluminescence detection system (GE Healthcare, USA) and analysed for band density using ImageJ densitometry software with GAPDH as a loading control.

2.15. Immunohistochemistry

Ileal and colonic tissues collected from mice were fixed in fresh 2% paraformaldehyde, and processed for H&E staining as previously described (Kozlowski et al., 2013). Slides were blindly scored by a pathologist for signs of histopathology and/or markers of inflammation and cellular infiltration.

2.16. Immunofluorescent microscopy

Cell monolayers grown on Lab-Tek chamber slides were washed twice with sterile PBS and fixed/permeabilized in cold methanol for 30 min at 4 °C. Non-specific binding was blocked with Heat inactivated faetal bovine serum (HI-FBS) for 15 min at room temperature and cells were incubated with primary antibodies for 1 h at 37 °C. In vivo sections of colonic tissues fixed in fresh paraformaldehyde, and embedded in paraffin, were cut into 5 µm segments and mounted on slides. After deparaffinization (three times, each 5 min: Neo-clear®, EMD Millipore, Billerica, MA, USA), sections were rehydrated in graded ethanol (100% twice,

95% twice, 70% twice, 5 min each) and were blocked/permeabilized in PBS/BSA/0.1%Triton (30 min; room temperature). Slides were incubated with primary antibodies (Thy-1.2/CD45R/TLR4 1:200) in a humid chamber at 4 °C (overnight) and rinsed (15 min) in PBS-T (room temperature). Secondary mAbs were administered (1:1000) for 1 h in PBS-T (room temperature), followed by a PBS wash and mounting.

2.17. Biofilm collection for germ-free mouse colonisation

CBD lids were removed from challenge plates and added to 96-well plates containing 200 µl/well of 0.09% saline, and sonicated on high for 10 min, Wells from six 96-well plates were pooled for each treatment group into 50 mL polypropylene Falcon tubes (VWR, USA), and centrifuged for 20 min at 1000*g*. The resulting pellet was then resuspended in 1.5 mL of sTSY and samples transported, overnight, to McMaster University (Hamilton, Ontario, Canada) where germ-free facilities are located.

2.18. Characterisation of microbiota biofilms exposed to G. duodenalis using Terminal restriction fragment Length Polymorphism (T-RFLP)

DNA was extracted from MBBF as well as planktonic constituents, using the methods previously outlined (Sproule-Willoughby et al., 2010). Extracted DNA was amplified using universal 16 s rRNA primers 8f(5′-VIC®-AGA GTT TGA TCC TGG CTC AG-3′) (Applied Biosystems) and 926r (5′-CCG TCA ATT CCT TTG AGT TT-3′) (Sigma). PCR mixtures were conducted in 50 µl volumes and contained 1 µl of DNA template, 5 µl of 10x PCR Buffer (without MgCl2; Invitrogen), 1.5 µl of MgCl2 (50 mM), 1 µl of dNTP mix (10 mM), 1 µl of forward primer (10 pmol/µl), 1 µl of reverse primer (10 pmol/µl), 0.25 µl of recombinant Taq polymerase and 39.25 µl of distiled deionised H_2O (all from QIAGEN, Germany). The reaction conditions were as follows: 94 °C for 2 min, 25 cycles of 94 °C, 56 °C and 72 °C for 1 min each, and a final extension at 72 °C for 10 min. PCRs were run in triplicate, pooled and purified using the QIAquick® PCR purification kit (QIAGEN). Samples were eluted in 30 µl of elution buffer. Purified PCR products were digested with *Hpa*II (Invitrogen) overnight at 37 °C. Digests were stopped by incubation at 65 °C for 15 min followed by further purification with the QIAquick® PCR purification kit and elution in 40 µl of elution buffer. Purified products were submitted to the University of Calgary Core DNA Services Laboratory (http://www.ucalgary.ca/dnalab/) for fragment analysis using an 3730xl (96 capillary) genetic analyzer (Applied Biosystems). The LIZ1200 size standard (Applied Biosystems) was used to size fragments together with the G5 filter set to detect VIC®-labelled fragments. T-RFLP data were analysed using GeneMapper 3.0 Software (Applied Biosystems). A virtual digest of the "H.Q. 16S Gut Organisms" Database of the restriction digest products (RDP) was conducted using the Microbial Community Analysis (MiCA) III program (Department of Biological Sciences, University of Idaho, USA http://mica.ibest.uidaho.edu/) as described previously (Sepehri et al., 2007; Sproule-Willoughby et al., 2010).

2.19. Biofilm administration into germ-free mice

Germ-free C57BL6 mice were housed at McMaster University (Hamilton, ON, Canada), in certified germ-free facilities. Over the course of a 2 week period, animals received two sets of biofilm communities via intra-rectal administration. Mice received biofilm contents that were exposed to either live *Giardia* trophozoites or control sTSY medium. Upon the first inoculation, animals were removed from the germ-free facilities. A true germ-free control group was maintained within the germ-free facility. Another inoculation was performed 7 days later to optimise the chances of colonisation. Following an additional 7 days, animals were sacrificed and analysed for gross morphology. Colonic and small intestinal samples were also collected for immunohistochemistry, western blot analysis, and reverse transcription (RT)–PCR) for microbial ecology analysis. Additionally, caecal swabs were collected to analyse the success of bacterial colonisation in the animals. Colon and faecal samples were assessed for bacterial species diversity by DNA analysis. All animal experiments were compliant with the animal ethics requirements of the University of Calgary and McMaster University, Canada and with the Canadian Council of Animal Care.

2.20. Multiplex LASER bead assay

Colonic segments collected from reconstituted germ-free mice were incubated in RIPA buffer (1× PBS, 1% Igepal CA-630, 0.5% sodium deoxycholate, and 0.1% SDS – all from Sigma) containing a protease inhibitor tablet (Complete-Mini, Roche Diagnostics), and homogenised with sterile 0.2 mm metal beads (BioSpec Products Inc., USA) and bead beating in a Precellys® 24 lysis homogenisation system (Bertin Technologies, Norway) (6500 rpm, 30 s beating, 45 s rest, three times). Homogenates were subsequently sonicated (Level 3, 5 s) and centrifuged at 10,000*g* for 10 min. Resulting supernatants were collected and protein concentration was determined using a Bradford Assay (Bio-Rad Laboratories). Colonic tissue homogenates were normalised to 4 mg/mL of protein and sent to undergo cytokine multiplex LASER bead array analysis via Eve Technologies Discovery Assay® (Calgary, AB, Canada).

2.21. DNA extraction

DNA from this faecal matter was extracted using a QIAGEN QIAamp DNA Stool Mini Kit. By contrast, DNA was extracted from planktonic cells and biofilms by established protocols (Sproule-Willoughby et al., 2010). Here, planktonic cells from three wells of the CBD (~450 µl) were collected by centrifugation at >10,000*g*, and pellets were suspended in 500 µl of lysozyme solution which contained 20 mg/mL of lysozyme (Sigma), 2.0 mM EDTA and 1.2% Triton X-100 made up in 20 mM Tris–HCl, pH 8.0. Alternatively, biofilms from eight pegs from the CBD were disrupted by sonication directly into lysozyme solution. Bacteria were lysed by 30 min incubation of bacteria at 37 °C in lysozyme solution,. Next, 500 µl of buffer ATL (QIAGEN) and 0.02 µl of Proteinase K (QIAGEN) were added with 0.5 g of sterile 0.1 mm glass beads (BioSpec Products) and samples underwent bead beating in a Precellys® 24 lysis homogenisation system (Bertin Technologies) (three cycles of 5000 rpm for 30 s with 45 s on ice). Buffer AL (QIAGEN) was added and samples were incubated at 70 °C for 10 min, followed by the addition of absolute ethanol to precipitate DNA. Finally, the QIAamp DNA Mini Kit (QIAGEN) was used for rapid purification of extracted DNA according to the manufacturer's directions except with two modifications: (i) each sample was incubated with 10 µl of RNAse A (10 mg/mL) at 37 °C for 30 min prior to ethanol extraction, and (ii) elution was performed in two 100 µl aliquots to increase DNA concentration. In all cases, DNA quantity and quality were determined by A_{260}/A_{280} measurements, and samples were normalised to 5 ng/µl for use in sequencing library construction.

2.21.1. 16S amplicon sequencing libraries and illumina® sequencing

16S amplicon sequencing libraries were prepared by two-step PCR. This was carried out according to the application note "Preparing 16S Ribosomal RNA Gene Amplicons for the Illumina Miseq® System" (Part # 15044223 Rev. B) provided by Illumina®.

Bacterial 16S rDNA genes were amplified using primers S-D-Bact-0341-b-S-17/S-D-Bact-0785-a-A-21 (Klindworth et al., 2013) in triplicate, pooled, cleaned-up with the QIAquick PCR Purification Kit (QIAGEN), and quantified using a Qubit 2.0 dsDNA HS Assay Kit (Life Technologies). PCR products were normalised to 5 ng/μl and then barcoded with oligonucleotides matching those from the Nextera XT Index Kit (96 indices, Illumina®). Successful amplification of the target 464 bp amplicon, which spans the V3 and V4 regions of the 16S rRNA gene, was verified by standard procedures for electrophoresis on 0.8% agarose TRIS-acetate-EDTA gels (Green and Sambrook, 2012). Paired-end 16S amplicons were quantified, normalised and pooled, and subsequently sequenced on the Illumina® MiSeq platform using a Miseq Reagent Kit V2 (500 – cycles). Multiplexed sequencing libraries were adjusted to a final concentration of 12 picomolar (pM), and contained 30% PhiX Control V3 library (Illumina).

2.22. Bioinformatic analysis of operational taxonomic units (OTUs)

OTUs were constructed using the forward read only due to low quality in the 3′ end of reads. Raw reads were processed using Cutadapt 1.8 with the options '-e 0 -O 17 -g CCTACGGGNGGCWGCAG -a GGATTAGATACCCBDGTAGTC', to remove the primer sequences and any preceding adaptors. Subsequent processing was done using the UPARSE pipeline (Edgar, 2013) as implemented in usearch 8.1.1861. The clipped reads were filtered with the usearch -fastq_filter using the options '-fastq_maxee 1.0 -fastq_ trunclen 150′. In addition to filtering, the clipped reads were converted to Fasta format for later use in constructing the OTU table. The combined filtered reads were dereplicated using the usearch -derep_fulllength command and the clustered with usearch -cluster_otus and the option '-minsize 2' to remove singleton reads prior to clustering. Taxonomy was assigned to the representative sequences using usearch -utax with the option '-utax_cutoff 0.8'. The taxonomy database was constructed with usearch -makeudb_utax using the RDP training set 15 as provided on the usearch website (http://drive5.com/usearch/manual/utax_downloads.html). The final OTU table was constructed with usearch -usearch_global and the options '-strand plus -id 0.97'. The entire procedure was run as a Snakemake pipeline (Koster and Rahmann, 2012).

2.23. Bacterial diversity analysis

Downstream analysis was done in R 3.2.2 (R.C. Team, 2015. A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, http://www.R-project.org) using phyloseq 1.14.0 (McMurdie and Holmes, 2013) and vegan 2.3–3 (Oksanen J, B.F., Kindt R, Legendre P, Minchin PR, O'Hara RB, et al, 2015. Vegan: Community Ecology Package. https://cran.r-project.org, https://github.com/vegandevs/vegan). β-Diversity was evaluated using the Bray-Curtis distance metric and visualised with non-metric multidimensional scaling (NMDS). To test if there was a difference between 'Control' and '*Giardia*' groups based on community distances, a permutational multi-variate ANOVA (PERMANOVA) was used as implemented in the adonis function of vegan. α-Diversity was measured with the Shannon index and differences in diversity between the Control and *Giardia* groups was tested with the Mann-Whitney test. OTUs that differed significantly between the Control and *Giardia* groups were identified with a generalised linear model approach implemented in DESeq2 (Love et al., 2014). This approach appropriately controls for over-dispersed data and variable library sizes (McMurdie and Holmes, 2014).

2.24. Standard molecular methods and reagents

All oligonucleotides, including primers and sequencing indices for sequencing library construction, were purchased from Integrated DNA Technologies, USA.

2.25. Statistical analysis

Results were expressed as means ± S.E.M. and compared by one-way ANOVA, followed by Tukey's test for multiple comparison analysis where needed. Statistical significance was established at $P < 0.05$.

2.26. Ethics statement

All studies involving human small intestinal mucosal biopsy tissues were approved by the Conjoint Health Research Ethics Board (CHREB) at the University of Calgary, Canada and the Calgary Health Region, Canada. In accordance with CHREB guidelines, adult subjects used in this study provided informed, written consent and a parent or guardian of any child participant provided informed, written consent on their behalf.

All animal experimentations were compliant with the animal ethics requirements of the University of Calgary, Canada and McMaster University, Canada, and with the Canadian Council of Animal Care. Certification of Animal Protocol Approval was provided by the Life and Environmental Sciences Animal Care Committee of the University of Calgary (ID # BI09R-05).

3. Results

3.1. Giardia changes the species composition of bacterial communities that colonise the intestine of germ-free mice

Although there are several animal models available for studying giardiasis (Roberts-Thomson et al., 1976; Stevens and Roberts-Thompson, 1978), we postulated that some of the key interactions between *Giardia* and intestinal microbiota might be modelled in the absence of the host. Thus, we cultured and enriched microbiota from human colon biopsies in laboratory biofilms using

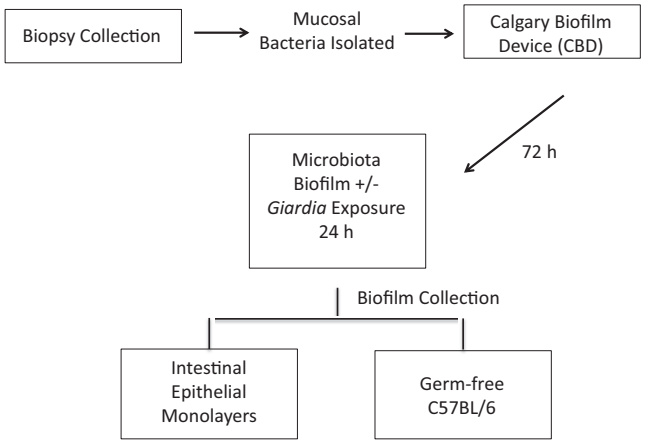


Fig. 1. Outline of experimental procedure. Human mucosal biopsies were collected from healthy donors (see Section 2.2), and mucosal bacteria were isolated and seeded into the Calgary Biofilm Device. Microbiota biofilms were cultivated on the Calgary Biofilm Device for 72 h and exposed to live *Giardia duodenalis*, spent medium, or control conditions for 24 h. Following exposure, biofilms were processed for microscopic analysis, intestinal epithelial monolayer challenge, or inoculation into germ-free C57BL/6 mice. Multiple aliquots of three biopsies, from a total of six donors, were used to seed biofilms used for all experiments.

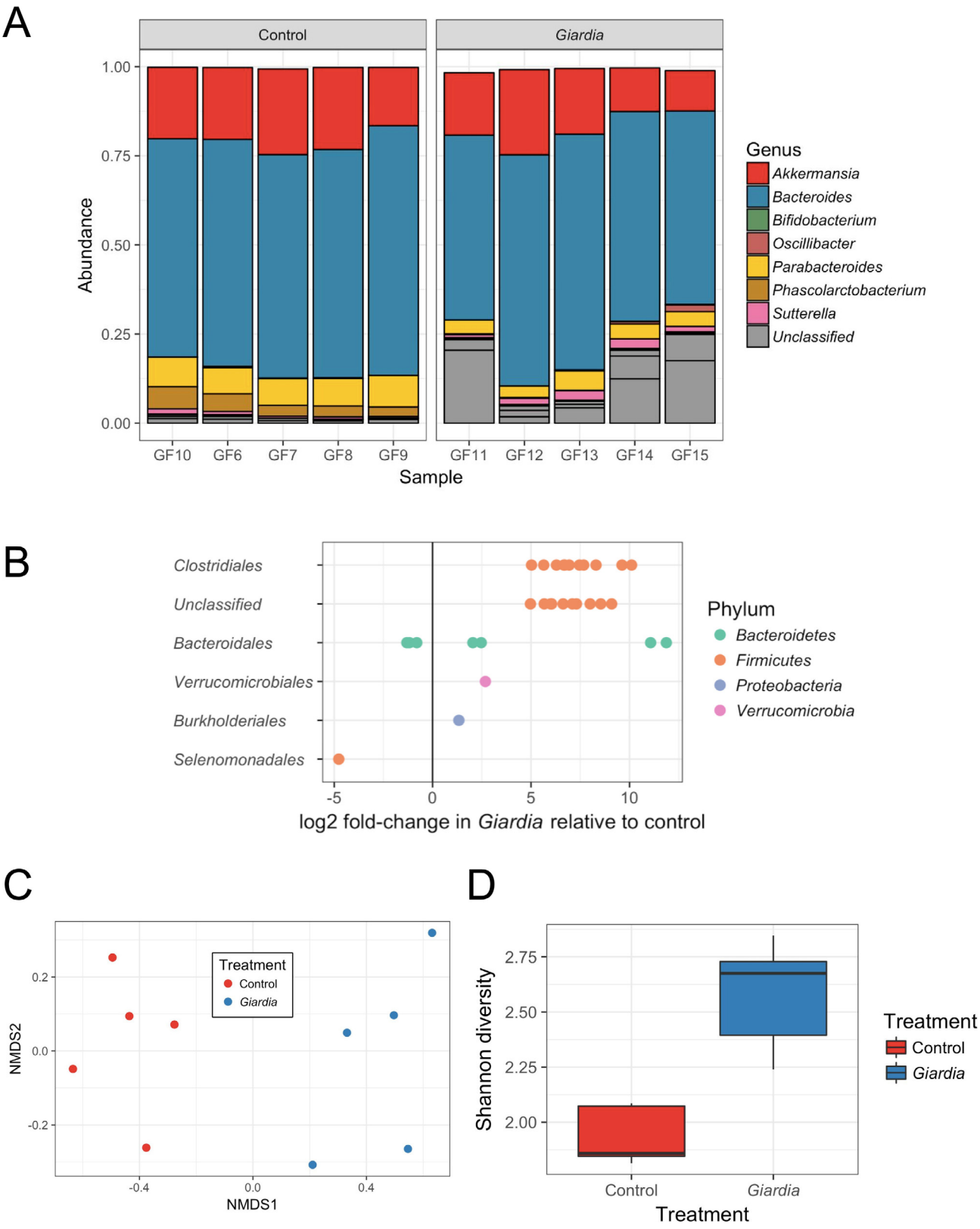


Fig. 2. Community profiles of faecal microbiota of germ-free mice reconstituted with human microbiota biofilms; mice were given either untreated microbiota (Control), or microbiota challenged with *Giardia duodenalis* (*Giardia*). Germ-free C57Bl6 mice were administered two rounds of human microbiota biofilm communities exposed to *G. duodenalis*, or control medium containing no parisites, and were euthanised after 14 days ($n = 5$ per group). Faecal samples were collected from each animal and analysed via 16S amplicon sequencing of the V3–V4 rRNA region. (A) Operational taxomomic units in faeces of mice either given untreated microbiota (Control), or microbiota challenged with *G. duodenalis* (*Giardia*). (B) Changes in operational taxomomic unit abundance by phylum in the same samples. (C) Non-metric multidimensional scaling ordination plot of β-diversity using the Bray–Curtis metric in the same samples. (D) α-Diversity of the same samples as assessed using the Shannon diversity index.

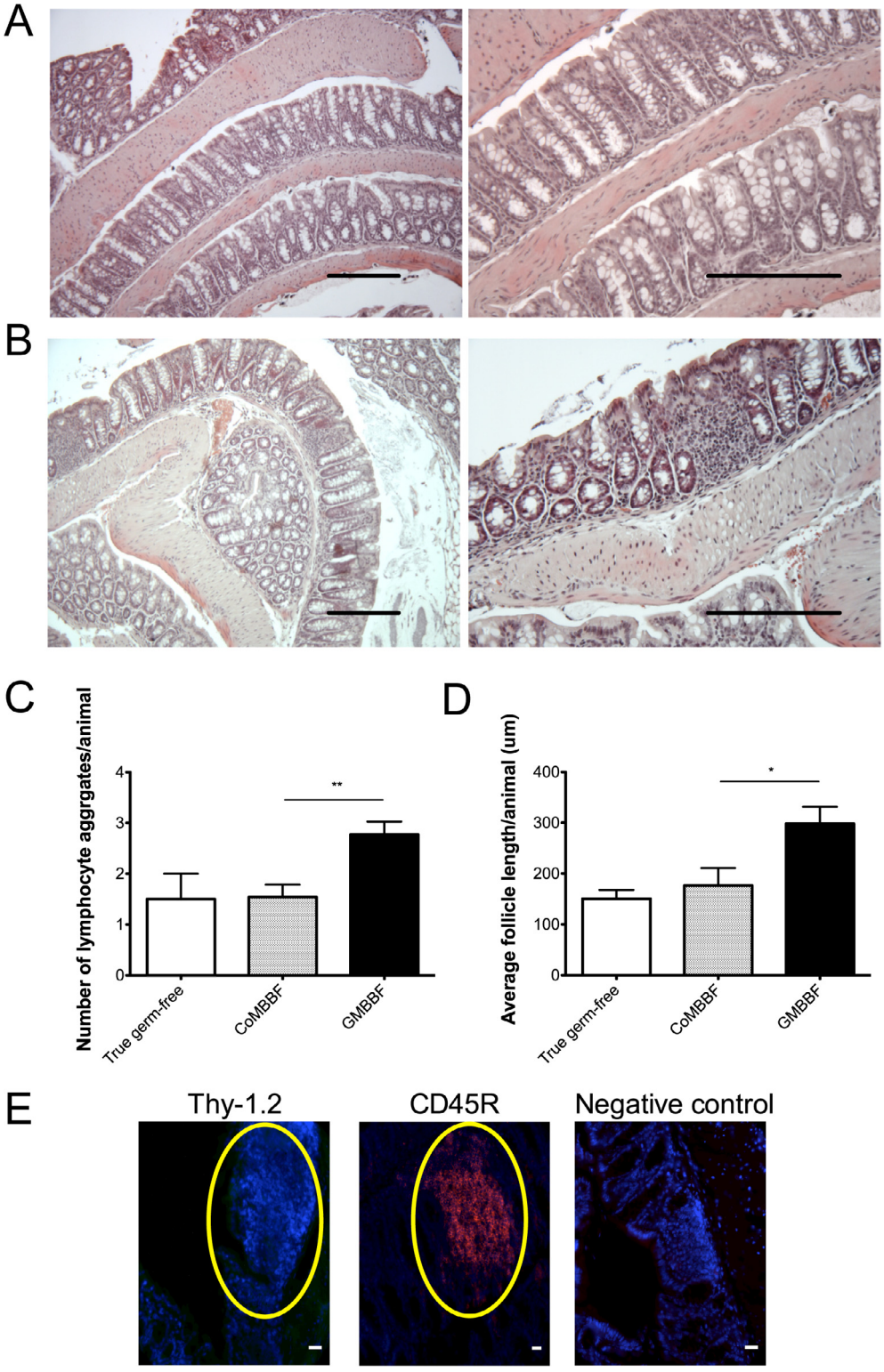


Fig. 3. *Giardia*-exposed microbiota biofilm communities promote B cell-containing lymphocyte follicles in the colonic epithelium of reconstituted germ-free mice. Sections of colon tissue from reconstituted germ-free C57Bl/6 mice. Animals were given intercolonic administration of two rounds of human microbiota biofilm communities exposed, or not, to *Giardia* and were euthanised after 14 days (n = 5–10 per group). Representative pictures of colon tissue stained with H&E from mice receiving either control biofilms CoMBBF (A), or those exposed to *Giardia* GMBBF (B). The average number (C) and average size (D) (recorded blindly) of colonic lymphocyte follicles per microscopic field (200X) in groups of mice receiving control microbiota biofilms (CoMBBF), and mice receiving *Giardia*-exposed microbiota biofilms (GMBBF) ($\geq$four different fields were analysed per animal. True germ-free animals were neither exposed to CoMBBF nor GMBBF, but represented a measure of baseline colonisation following transfer from germ-free facilities to the environment where experimentation took place. (E) Immunohistochemistry was performed on colonic sections of reconstituted germ-free mice. Representative images of animals receiving *Giardia*-exposed microbiota biofilms are presented as stained for T and B cell phenotypes. Host nuclei were coloured in blue (DAPI), Thy-1.2 (CD90-pan T cell marker), and CD45r-positive cells in red (rat polyclonal anti-CD45R coupled with alexa455). Lymphocyte aggregates were negative for Thy-1.2 (CD90-pan T cell marker) staining. Negative control staining used rat IgG coupled with the same fluorochrome. Scale bars represent 50 μm for E and represent the limit of intraepithelial lymphocyte follicles ($n \geq 4$ different fields were analysed per animal). $* P < 0.05$, $** P < 0.01$ (ANOVA and Tukey's multiple comparison test for data in C and D).

microaerobic conditions in the CBD (Sproule-Willoughby et al., 2010). An initial set of experiments had determined whether *G. duodenalis* was able to directly alter human microbiota, using T-RFLP analysis prior to introducing the microbiota into germ-free mice (Supplementary Fig. S2). The data illustrated that exposure to *Giardia* directly alters human microbiota with an apparent over-representation of *Clostridiales* bacteria. Subsequently, these MBBFs were exposed to live *Giardia* or a sterile growth medium. *Giardia*-treated and control biofilm communities were then disrupted and administered to two groups of germ-free mice (this experimental design is illustrated in Fig. 1). A third group of mice was administered a sham control treatment. Experiments were performed using both live *Giardia* and conditioned medium, in order to determine parasite impact on biofilm communities in the presence/absence of physical interference induced by live enteropathogens.

All mice receiving MBBFs showed a decrease in caecum weight relative to the sham controls, suggesting successful bacterial colonisation of the murine gastrointestinal tract Supplementary Fig. S3. We evaluated the species composition of intestinal microbiota communities in mice receiving *Giardia*-treated biofilms (GMBBFs) and control MBBF groups by 16S amplicon sequencing on the Illumina® platform. Note that caecal swabs taken from each animal were evaluated for bacterial growth on agar medium, and animals from the sham control group were culture negative at all times. DNA from the faeces of the sham control group did not yield 16S amplicons by PCR.

Classification of 16S amplicon sequences (from 5,906 to 55,526 clusters passing filter for each replicate, Supplementary Table S1) was carried out using the UPARSE pipeline (Fig. 2A, see Sections 2.21–2.25). The most striking changes in mice receiving *Giardia*-treated microbiota were increases in genera belonging to the order Clostridiales (Firmicutes), and a decrease in bacteria belonging to the genus *Phascolarctobacterium* (Fig. 2B, Supplementary Table S2). β-Diversity was assessed using the Bray–Curtis distance metric, and while samples showed large within group variation, the largest source of variation between groups was treatment with *Giardia* (Fig. 2C). α-diversity was measured using the Shannon diversity index, and microbiota in faeces from mice receiving *Giardia*-treated biofilms had greater α-diversity than those mice receiving control biofilms (Fig. 2D, $P < 0.01$ by Mann–Whitney *U*-test).

3.2. Giardia-modified biofilms promote the formation of mucosal B lymphocyte follicles and increase the production of pro-inflammatory cytokines

Histological examination of colonic morphology in mice receiving human microbiota biofilms revealed lymphocyte aggregations within the colonic mucosa of animals reconstituted with GMBBF communities (Fig. 3A and B). Upon blind scoring, the number and size of the lymphocyte follicles were determined to be more numerous and larger, respectively, in animals receiving GMBBF communities (Fig. 3C and D). Immunohistochemistry revealed the lymphocyte follicles to be encompassed primarily by B lymphocyte populations and not T cells (Fig. 3E).

Another set of studies examined levels of important pro-inflammatory mediators, IL-6, IL-1β and TNF-α, as well as IFN-γ (Supplementary Table S1). IL-1β levels were significantly increased in colonic tissues from animals receiving human microbiota modified by exposure to *Giardia* (Fig. 4). In these tissues, IL-6 (Control = 75 ± 9 pg/mL; *Giardia* = 515 ± 391 pg/mL), TNF-α, (Control = 35 ± 5 pg/mL; *Giardia* = 188 ± 96 pg/mL), IFN-γ (Control = 51 ± 6 pg/mL; *Giardia* = 752 ± 614 pg/mL), and IL-17 (Control 98 ± 60 pg/mL; *Giardia* = 467 ± 274 pg/mL), saw their mean concentrations raised by more than 8, 5, 14 and 4-fold, respectively (Supplementary Table S1).

3.3. Giardia reduces the thickness of human microbiota biofilms ex vivo and alters the biofilm extracellular matrix

In an attempt to further characterise the alterations to microbiota biofilms induced by exposure to *G. duodenalis*, we assessed changes in phenotype and bacterial composition of the bacterial communities. Confocal scanning laser microscopy images were taken from the CBD grown biofilms at the atmosphere-liquid interface, where biofilm formation was optimal, and where determination of thickness could be standardised across experiments. The observations indicate that exposure to *Giardia* reduces the thickness of human microbiota biofilms (10–105 μM) versus untreated controls (100–210 μm) (Supplementary Fig. S4). SEM revealed that the extracellular coating of the microbiota biofilms was altered or diminished by exposure to *Giardia*. WGA assays of the biofilm communities confirmed that *Giardia* significantly altered the chemical composition of the biofilm ECM, while exposure to a non-pathogenic microbial challenge (*E. coli* HB101) did not have this effect (Fig. 5B). Fluorescently-labelled WGA was used to detect N-acetylglucosamine residues, which are encompassed within the biofilm ECM of multiple bacteria (Perez-Mendoza et al., 2011), and represents an important surrogate to determine the presence of ECM in microbiota biofilm communities.

As the structural integrity of gastrointestinal biofilm communities depends on resident microbial cell production of ECM materials, we therefore aimed to determine a potential role for *Giardia* enzymatic products in the modification of the biofilm matrix. *Giardia* spent medium contained significant amounts of cysteine protease activity (Supplementary Fig. S5). The broad spectrum cysteine protease inhibitor E-64 significantly reduced cysteine protease activity in biofilm communities exposed to *Giardia* spent medium (Supplementary Fig. S5). To further characterise the effects of the *Giardia* proteases, specifically in the context of biofilm ECM, fluorescence-labelled WGA (revealing biofilm ECM) was assessed by confocal scanning laser microscopy. The observations suggest that *Giardia* spent medium alters the microbiota biofilm ECM in E-64-sensitive fashion (Fig. 5D). Biochemical measurements of WGA confirmed the visual observations (Fig. 5C). Conversely, CA-074Me, a Cathepsin-B specific inhibitor, was unable

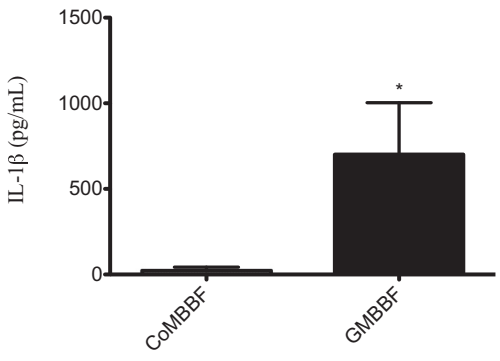
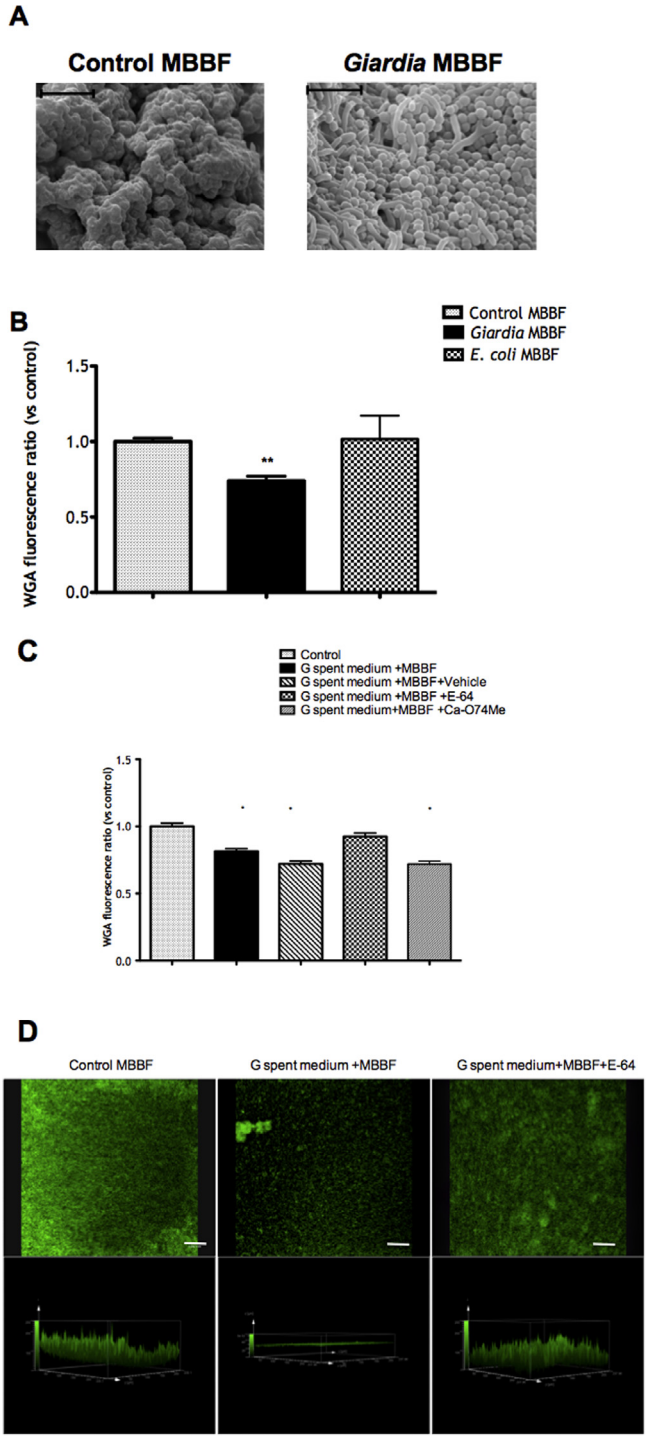


Fig. 4. Germ-free mice receiving *Giardia*-exposed microbiota biofilm communities (GMBBF) show increased colonic IL-1β protein levels. Animals were given intracolonic administration of two rounds of human microbiota biofilm communities exposed, or not, to *Giardia* and were euthanised after 14 days ($n = 4–5$ per group). Protein expression of IL-1β was measured in colonic segments collected from germ-free C57Bl/6 mice reconstituted with differential microbiota biofilm communities using the multiplex technology of the Eve Technologies Luminex platform. Mice receiving *Giardia*-exposed microbiota biofilm communities exhibited greater expression of IL-1β compared with those receiving control microbiota biofilms (CoMBBF). $^{*}P < 0.05$ (Student's *t*-test).

to reverse the detrimental effects of *Giardia* spent medium on the microbiota biofilm ECM, suggesting this particular enzyme was not responsible for inciting ECM changes in our model (Fig. 5C).

3.4. Bacteria released from the Giardia-modified microbiota biofilms promote enterocyte apoptosis and increase trans-epithelial permeability

In another set of studies we aimed to characterise the potential pathogenic effects of the planktonic bacteria released from *Giardia*-altered biofilms on human epithelial enterocytes in vitro. We first assessed epithelial apoptosis, an established marker of gut pathology in a variety of intestinal infections and disorders (Buret, 2007). Planktonic bacteria from biofilms exposed to *Giardia* spent medium significantly increased the levels of enterocyte apoptosis versus those exposed to control biofilms (Fig. 6). Additionally, bacteria released from microbiota biofilms exposed to *Giardia* translocated across epithelial monolayers at significantly higher levels than controls (Fig. 7). Consistent with the changes seen in the ECM, E-64 treatment of *Giardia* prevents translocation of the resulting planktonic bacteria (Fig. 7), suggesting that ECM integrity is necessary to prevent bacterial translocation by microbiota biofilm constituent bacteria.



3.5. Giardia-modified microbiota biofilm communities disrupt tight junctional ZO-1 and promote epithelial TLR4 expression and CXCL-8 secretion

The human intestinal epithelium must coexist with billions of commensal microbes, and even in the absence of pathogenic insult can secrete numerous inflammatory cytokines, chemokines and antimicrobial peptides. The formation and maintenance of tight junctions is crucial in coordinating the homoeostatic barrier function (O'Neil et al., 1999; Vora et al., 2004; Sawada, 2013), as is epithelial signalling through TLR4, particularly through regulating pro-inflammatory CXCL-8 secretion. In an attempt to further identify the mechanisms of epithelial pathophysiology induced by enteropathogen-modified microbiota biofilms, we then assessed the consequences of exposure to *Giardia*-modified microbiota constituents on tight junctional ZO-1, on TLR4 expression, and on the production of pro-inflammatory CXCL-8.

Epithelial monolayers incubated with planktonic communities from control microbiota biofilms exhibited normal membrane distribution of ZO-1 (Fig. 8B). In contrast, monolayers incubated with planktonic bacteria from biofilms exposed to *Giardia* spent medium showed reduction in tight junctional ZO-1 expression (Fig. 8B). These changes were accompanied by a significant reduc-

Fig. 5. *Giardia duodenalis* modifies human microbiota biofilm extracellular matrix structure, mediated in part by cysteine proteases. Human microbiota biofilm communities were cultivated on the Calgary Biofilm Device using colonic biopsies collected from healthy human donors. Following a 72 h growth period, microbiota biofilms were subject to either control conditions (fresh sTSY medium), *Escherichia coli*, exposure to *Giardia*, or exposure to *Giardia* (G) spent medium. (A) Representative Scanning Electron Micrographs of microbiota biofilms exposed to fresh sTSY medium (Control MBBF) or live *Giardia* trophozoites (*Giardia* MBBF). 13,000 × magnification. Scale bar = 5 μm. (B) Spectrofluorometric assay using Wheat-germ Agglutinin dye for the quantification of extracellular matrix in control microbiota mucosal biofilms (Control MBBF), those exposed to live *Giardia* trophozoites (*Giardia* MBBF), and those exposed to live commensal *Escherichia coli* (*E. coli* MBBF). WGA selectively binds to *N*-acetylglucosamine residues in the biofilm extracellular matrix. The data illustrate that *Giardia* reduces the extra-cellular exopolysaccharide matrix of human intestinal microbiota biofilms, an effect not seen when the biofilms were exposed to non-pathogenic *Escherichia coli* (B). (C) Spectrofluorometric assay using Wheat-germ Agglutinin dye for the quantification of extracellular matrix in microbiota biofilms exposed to fresh sTSY (Control), or *Giardia* spent medium (G spent medium MBBF), in the presence or absence of E-64, a membrane impermeable broad-spectrum cysteine protease inhibitor (G spent medium + MBBF + E-64). DMSO served as the vehicle control. *Giardia* spent medium-mediated breakdown in microbiota biofilm extracellular matrix is inhibited by pre-treatment with E-64. Ca-074Me, a Cathepsin B inhibitor, is unable to reverse the detrimental effect of *Giardia* spent medium on biofilm extracellular matrix integrity. (D) Representative confocal scanning laser micrographs of microbiota biofilms stained with WGA. Stain intensity, a marker of biofilm thickness, is lost upon exposure to *Giardia* spent medium (G spent medium + MBBF), and the effect is inhibited by pre-treatment with E-64 ($n \geq 3$ different fields were analysed per experiment, scale bar represents 106 cm). Values were calculated as fluorescence ratios versus values measured in control MBBF communities, arbitrarily set to 1.0. Values are mean ± S.E.M. $n = 3$/group. $^{**}P < 0.005$ versus Control MBBF (ANOVA and Tukey's multiple comparison test).

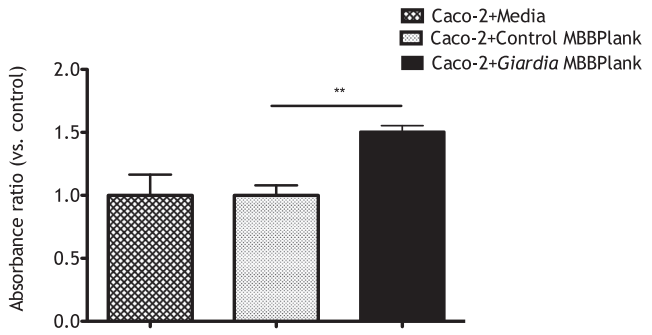


Fig. 6. Planktonic bacteria from human microbiota biofilms exposed to *Giardia* spent medium promote heightened rates of epithelial apoptosis. Human microbiota biofilm communities were cultivated on the Calgary Biofilm Device using colonic biopsies collected from healthy human donors. Following a 72 h growth period, microbiota biofilms were subjected to either control conditions (fresh sTSY medium) or exposure to *Giardia* spent medium. Caco-2 monolayers co-incubated, for 24 h, with planktonic constituents of microbiota biofilms exposed to *Giardia* spent medium (Caco-2 + Giardia MBBPlank) exhibited significantly higher rates of apoptosis compared with cells incubated with microbiota biofilm communities exposed to fresh sTSY (Caco-2 + Control MBBPlank). Caco-2 cells incubated with fresh cell medium served as baseline indicators of cell apoptosis. Values were calculated as absorbance ratios versus control: the mean of all control values was calculated and set at 1.0, and ratio changes were calculated for each measurement in each group against this calculated control mean. Values are mean ± S.E.M. $n = 3$/group. $^{*}P < 0.05$ versus Caco-2 + Medium (ANOVA and Tukey's multiple comparison test).

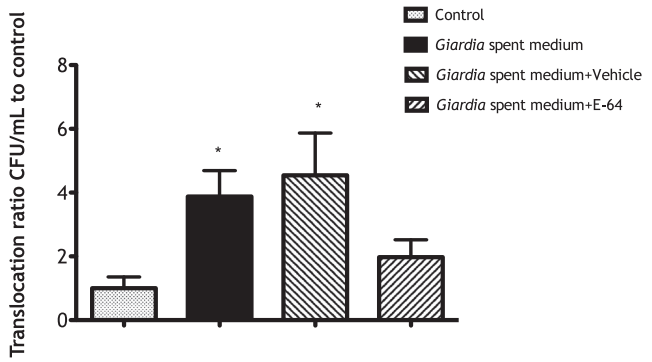


Fig. 7. Planktonic bacteria from human microbiota biofilms exposed to *Giardia* spent medium more readily translocate across epithelial monolayers. Human microbiota biofilm communities were cultivated on the Calgary Biofilm Device using colonic biopsies collected from healthy human donors. Following a 72 h growth period, microbiota biofilms were subjected to either control conditions (fresh sTSY medium), or exposure to *Giardia* spent medium. Bacteria released from human microbiota biofilm communities exposed to *Giardia* spent medium translocate at significantly higher levels than those exposed to fresh sTSY (Control), and this effect was inhibited by pre-treatment with E-64 (*Giardia* spent medium + E-64). DMSO pre-treatment served as the E-64 vehicle control. Values were calculated as log colony forming unit (CFU)/mL ratios versus Control CFU/mL, arbitrarily set to 1.0. Values are mean ± S.E.M. $n = 3$/group. $^{*}P < 0.05$ versus Control (ANOVA and Tukey's multiple comparison test).

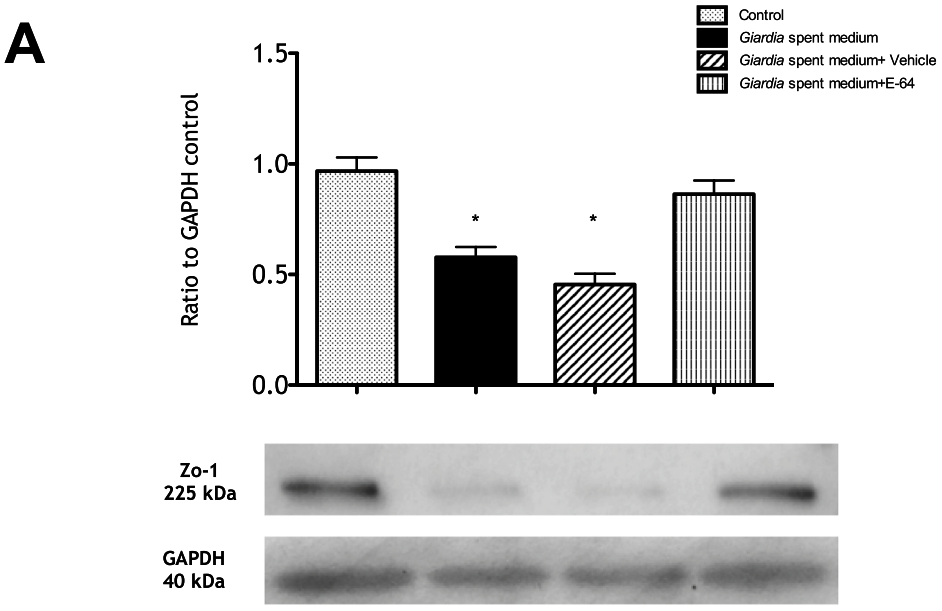
tion in the levels of ZO-1 total protein (Fig. 8A). E-64 pre-treatment of *Giardia* prior to spent medium preparation and subsequent incubation with microbiota biofilms prevented the abnormalities induced by the resulting planktonic bacterial constituents (Fig. 8). Furthermore, planktonic bacteria released following exposure of microbiota biofilms to *Giardia* spent medium significantly increased epithelial TLR4 expression, and this effect was dependent on cysteine protease activity (Fig. 9). Finally, planktonic bacteria from biofilms exposed to *Giardia* spent medium significantly induced increased epithelial CXCL-8 production, an effect that was also cysteine protease-dependent (Fig. 10).

4. Discussion

Based on our current study, we propose that *Giardia* may alter the structure and composition of human intestinal microbiota biofilms, and that bacteria from these dysbiotic microbiota in turn can cause epithelial and intestinal abnormalities after the enteropathogen has been cleared. Indeed, using a living host such as a germ-free mouse reconstitution model to look at the effect of different microbiome communities offers a powerful model to characterise potential pathological consequences of a *Giardia*-exposed community, in the absence of confounding parameters such as antimicrobials often used to clear gut microbiota. Germ-free mice receiving human microbiota modified by *Giardia* showed distinct increases in colonic IL-1β. Average levels of IL-6, TNF-α, IL-17 and IFN-γ were also several fold higher in comparison with mice that received control microbiota. Together, these findings support the hypothesis that a *Giardia*-perturbed microbiota community alone can promote pro-inflammatory profiles in the mammalian intestine. Moreover, we observed an increased number of lymphocyte follicles in mice receiving GMBBF, and these were constituted primarily of CD45 + B lymphocytes (Fig. 3E). Previous reports implicated augmented numbers of lymphocytes in intestinal pathology, notably in patients experiencing IBS (Spiller et al., 2000; Chadwick et al., 2002; Tornblom et al., 2002; Cremon et al., 2009; Ohman et al., 2009). Alterations in B cell function have been associated with mucosal inflammatory conditions, including in IBD (Noronha et al., 2009). There is evidence of sustained heightened expression of pro-inflammatory mediators, namely IL-1β mRNA and increased plasma levels of pro-inflammatory IL-6, TNF-α and CXCL-8, in samples collected from IBS patients, while other reports suggest a potential role for low-grade inflammation in IBD and IBS pathogenicity (Gwee et al., 2003; Liebregts et al., 2007; Scully et al., 2010). The specific roles for cytokines in IBS pathophysiology are not fully understood.

Microbiota dysbiosis, mostly based on faecal analyses, has been associated with a number of intestinal disorders including IBS. Distinct changes, at the phylum level, in the relative number of *Firmicutes* and *Actinobacteria* have been documented in IBS patients (Kassinen et al., 2007). T-RFLP analyses revealed an increase of Clostridiales bacteria in human microbiota biofilms exposed to *Giardia* ex vivo, suggesting a direct impact of the parasite on community composition. The present study also found a relative expansion in the abundance of Firmicutes (which Clostridiales are part of) and Bacterioidetes in the faeces of germ-free mice administered *Giardia*-treated microbiota communities.

Within the human gastrointestinal tract, the mucosa exists in close association with biofilms containing the commensal microbiota (Kleessen and Blaut, 2005). Intrinsic to the ability of bacteria to persist in biofilms is their coordinated secretion of substances constituting an ECM, of which polysaccharides and proteins are the important components (Sutherland, 2001). Exposure of human microbiota biofilms to *Giardia* and *Giardia* spent medium, resulted in a significant reduction in ECM. *Giardia* trophozoites contain and secrete copious amounts of proteolytic proteins (Hare et al., 1989; Williams and Coombs, 1995; David et al., 2012), and the recently sequenced *Giardia* genome indicates the presence of at least 23 genes encoding for cathepsin-like cysteine proteases of largely unknown function (DuBois et al., 2006). *Giardia* cathepsin proteases are thought to play a role in its encystation and excystation, as well as nutrient metabolism (DuBois et al., 2006). Recent findings also indicate that *Giardia* cathepsins contribute to immunomodulation and pathogenesis by cleaving pro-inflammatory mediators in the intestinal mucosa and disrupting epithelial cytoskeletal protein villin, respectively (Cotton et al., 2014; Bhargava et al., 2015; Cotton et al., 2015). Bacterial biofilm ECM is diverse in composition and



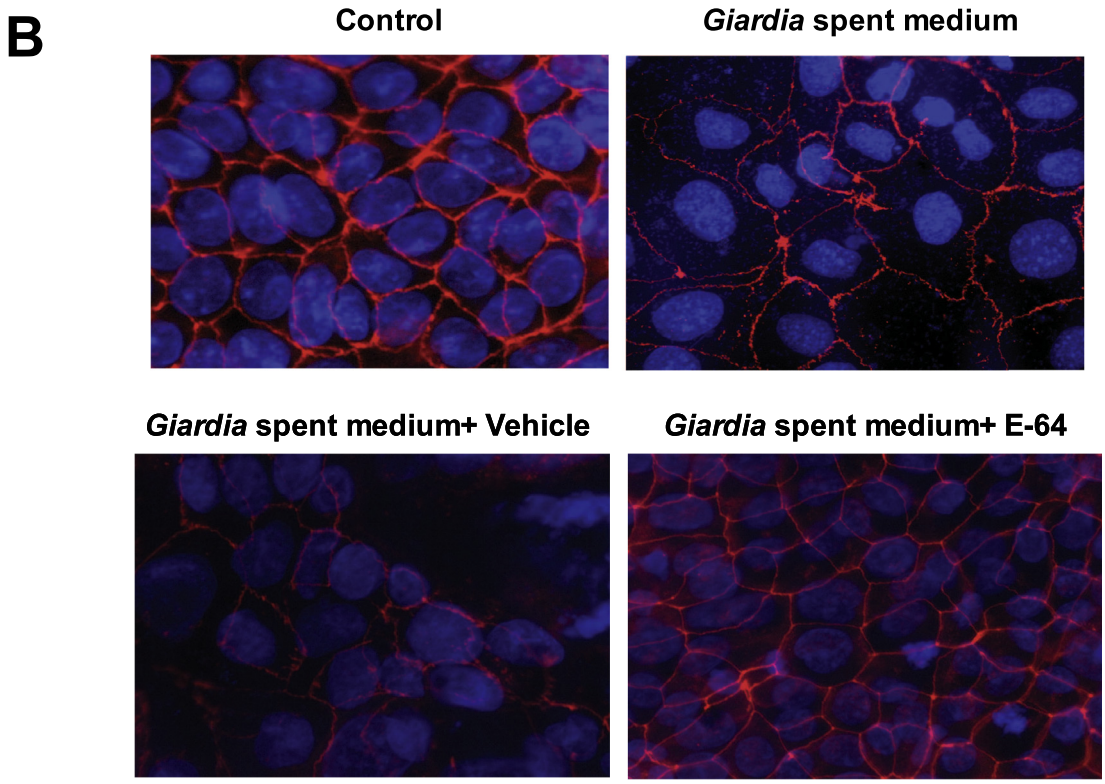


Fig. 8. Planktonic bacteria from human microbiota biofilms exposed to *Giardia* spent medium reduce zonula occludens (ZO)-1 protein and modify ZO-1 distribution in Caco-2 epithelial monolayers. Human microbiota biofilm communities were cultivated on the Calgary Biofilm Device using colonic biopsies collected from healthy human donors. Following a 72 h growth period, microbiota biofilms were subjected to either control conditions (fresh sTSY medium), or exposure to *Giardia* spent medium. Cell monolayers were challenged with planktonic constituents released from microbiota biofilms, either exposed to *Giardia* spent medium or fresh sTSY (Control) in the presence/absence of broad-spectrum, E-64, pre-treatment on trophozoites, for 24 h, and immunoblotted for ZO-1 (A). Western blots (densitometry and representative blots images) illustrate that Caco-2 monolayers co-incubated with planktonic constituents released from *Giardia* spent medium- exposed microbiota biofilm communities contain significantly decreased ZO-1 protein levels. Pre-treatment of trophozoites with E-64 inhibited this decrease. (B) Representative immunostaining for ZO-1 (red) reveals a mislocalization, and loss in stain intensity in Caco-2 monolayers co-incubated with planktonic constituents released from microbiota biofilm communities modified by exposure to *Giardia* spent medium. Densitometry data are presented as % glyceraldehyde 3 phosphate dehydrogenase. Values were calculated as % glyceraldehyde 3 phosphate dehydrogenase ratios versus control, arbitrarily set to 1.0 (A). Values are mean ± S.E.M. $n = 3$/group. $^{*}P < 0.05$ versus Control (ANOVA and Tukey's multiple comparison test).

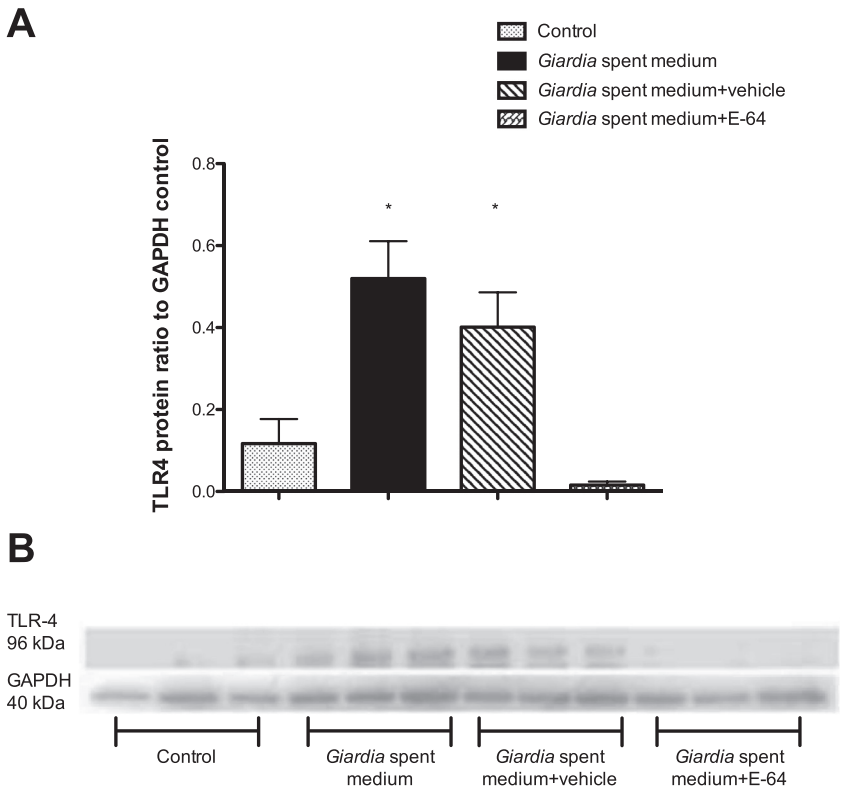


Fig. 9. Planktonic bacteria from human microbiota biofilms exposed to *Giardia* spent medium increase Toll-like receptor 4 expression in Caco-2 monolayers. Human microbiota biofilm communities were cultivated on the Calgary Biofilm Device using colonic biopsies collected from healthy human donors. Following a 72 h growth period, microbiota biofilms were subjected to either control conditions (fresh sTSY medium), and either control conditions (Control MBBPlank whereby no exposure to *Giardia* occured; or fresh sTSY medium), or exposure to *Giardia* spent medium (where trophozoites had been pre-treated with cysteine protease inhibitor E-64 or not), or exposure to the known pro-inflammatory cytokine IL1 beta. Data were also obtained from plain, untreated, CaCo-2 monolayers (Caco-2). (A) Caco-2 monolayers co-incubated with planktonic constituents released from human microbiota biofilms exposed to *Giardia* spent medium show a significant increase in TLR4 protein levels. Pre-treatment of trophozoites with E-64 prevented this increase of TLR4 protein levels in the monolayers. Densitometry data are presented as % glyceraldehyde 3 phosphate dehydrogenase. Values were calculated as % glyceraldehyde 3 phosphate dehydrogenase ratios versus control, arbitrarily set to 1.0. Values are mean ± S.E.M. $n = 3$/group. $^{*}P < 0.05$ versus Control (ANOVA and Tukey's multiple comparison test). Numbers were representative of separate experiments conducted with microbiota biofilm cultures cultivated from colonic samples of multiple patients in order to ensure reproducibility. (B) Caco-2 monolayers co-incubated with planktonic constituents released from human microbiota biofilms exposed to *Giardia* spent medium show a significant increase in secreted CXCL-8. To further identify the mechanisms of epithelial pathophysiology induced by enteropathogen-modified MBBFs, we assessed the consequences of exposure to *Giardia*–modified microbiota constituents on the production of pro-inflammatory CXCL-8. Pre-treatment of trophozoites with E-64 inhibited this increase. Values were calculated as % control arbitrarily set to 1.0. Values are mean ± S.E.M. $n = 3$/group. $*P < 0.05$ versus Control (ANOVA and Tukey's multiple comparison test).

proteins constitute a significant portion of the matrix (Toyofuku et al., 2012). Studies need to assess whether bacterial biofilm ECM associated protein (bap), pili, or fimbriae, among others, may provide substantial substrate possibilities to promiscuous cathepsins, such as those found in *Giardia*.

The present study demonstrates that *Giardia*, via its secretory-excretory proteases, alters microbiota biofilms. Planktonic bacteria released from these biofilms caused epithelial abnormalities. Planktonic bacteria released from biofilms may exhibit increased virulence (Chua et al., 2014). Moreover, it has been recently established that upon yet unknown stimuli, certain intestinal commensal organisms may behave as "pathobionts" that have the potential to initiate an aberrant host immune response, and to favour a pro-inflammatory state within the intestine (Round and Mazmanian, 2009). *Campylobacter jejuni*, another enteropathogen known to cause PI-IBS, is able to promote the trans-epithelial translocation of non-invasive commensal bacteria by hijacking host lipid rafts (Kalischuk et al., 2009), and recent findings showed that exposure to *C. jejuni* may activate latent virulence genes in commensal *E. coli* (Reti et al., 2015). The present findings indicate that planktonic bacteria released from the *Giardia*-treated microbiota biofilms induced apoptosis and pro-inflammatory cytokine release in enterocytes. The observations are consistent with the recent findings that human microbiota exposed to *Giardia* cause lethal paralysis in *Caenorhabditis elegans* (Gerbaba et al., 2015). Further research will determine whether these observations may together offer one explanation of how certain microbiota commensals may become pathobionts. Indeed, heightened enterocyte apoptosis represents a key feature of various intestinal disorders (Ramachandran et al., 2000; Yu et al., 2005; Buret and Bhargava, 2014). Moreover, the findings described here also show that planktonic bacteria from dysbotic microbiota readily translocate across epithelial monolayers. More research will assess how these observations may offer another hypothesis to explain the mechanisms underlying post-infectious abnormalities such as IBS (Barbara et al., 2002; Tornblom et al., 2002; Mason et al., 2011). This enhanced translocation, as well as tight junctional ZO-1 disruptions, were prevented when *Giardia* secretory-excretory proteases were inhibited. The proper maintenance of tight junctions is crucial in coordinating homoeostatic barrier function within the gastrointestinal tract, and preventing increased permeability often associated with disease progression (Sawada, 2013). Indeed, an intriguing avenue to explore in future studies would be to examine the impact of *Giardia*-treated human microbiota communities on intestinal permeability in vivo, using the reconstituted germ-free mouse model described here.

Numerous reports suggest a role for dysregulated intestinal immunity in the pathogenesis of IBS as well as IBD (Konig and

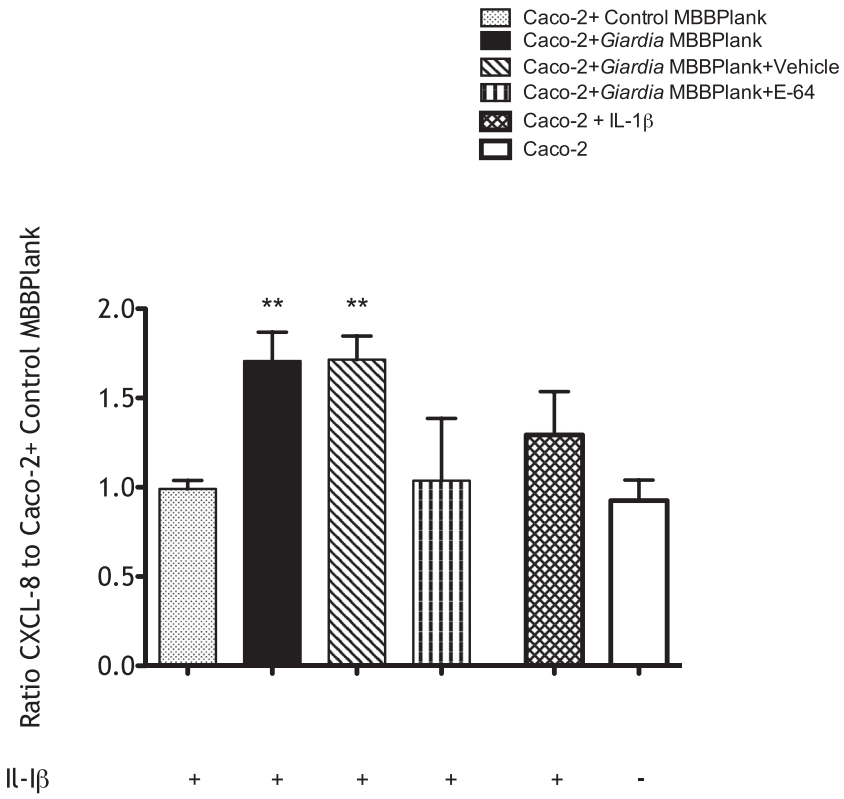


Fig. 10. *Giardia* spent medium-exposed microbiota biofilms planktonic constituents (MBBPlank) promote increased CXCL-8 protein levels secreted from Caco-2 monolayers. Human microbiota biofilm communities were cultivated on the Calgary Biofilm Device using colonic biopsies collected from healthy human donors. Following a 72 h growth period, microbiota biofilms were subjected to either control conditions (fresh sTSY medium), or exposure to *Giardia* spent medium. Caco-2 monolayers co-incubated with planktonic constituents released from *Giardia* spent medium-exposed microbiota biofilm communities showed a significant increase in secreted CXCL-8. Pre-treatment of trophozoites with cysteine protease inhibitor E-64 prevented increased secreted CXCL-8 levels in stimulated Caco-2 monolayers challenged with planktonic microbial constituents released from microbiota biofilms. Values were calculated as % control, arbitrarily set to 1.0. Values are mean ± S.E.M. $n = 3$/group. $^{*}P < 0.05$, $^{**}P < 0.01$ versus Control (ANOVA and Tukey's multiple comparison test). Cells given IL-1 beta served as a positive "pro-inflammatory" control.

Brummer, 2014). ZO-1 is a particularly important epithelial scaffolding constituent allowing for signal transduction that is vital to the dynamic nature of tight junction formation and function (Sawada, 2013) Epithelial lipopolysaccharide (LPS) signalling at least in part regulates CXCL-8, secretion from intestinal epithelial cells through TLR4 (O'Neil et al., 1999; Vora et al., 2004), and increased TLR4 levels have been reported in inflammatory disease states such as in IBD (Cario and Podolsky, 2000; Abreu et al., 2002) and IBS (Belmonte et al., 2012). Similarly, while critical to the prevention of microbial invasion and infection, abnormally high levels of CXCL-8 are known to be present in perpetuated inflammatory conditions of the gut (O'Neil et al., 1999; Fahlgren et al., 2003; Wang et al., 2009).

CXCL-8 secretion can serve as an important functional determinant of intestinal epithelial cell responses to bacterial products due to its integral involvement in the recruitment of immune cells to sites of bacterial infection and inflammation (Chen et al., 2009). Strikingly, epithelial CXCL-8 release was significantly increased when enterocytes were exposed to planktonic bacteria from *Giardia*-treated microbiota biofilms. Future research will determine whether this may help explain, at least in part, post-infectious infiltration of neutrophils in giardiasis. These abnormalities coincided with increased epithelial TLR4 protein levels. Basal homoeostatic epithelial cell expression of TLR4 is low, creating an LPS-unresponsive intestinal milieu; this TLR4 expression is significantly increased in inflammatory disease states, such as IBD and IBS (Cario and Podolsky, 2000; Abreu et al., 2002). The data reported herein offers a novel hypothesis whereby *Giardia*-perturbed microbiota biofilm communities may promote epithelial expression of TLR4, which in turn may facilitate pro-inflammatory cytokine secretion down-stream.

Taken together, the present findings reveal new mechanisms causing sustained inflammatory mechanisms that surround post-infectious intestinal disorders, especially in the context of enteropathogen-induced altered microbiota dynamics. Moreover, the findings shed new light on the key significance of the polymicrobial interplay that occurs between an enteropathogen and the commensal microbiota community. Further understanding of the mechanisms through which a pathogen such as *Giardia* incites microbiotic changes, and of how these changes persist within the gastrointestinal tract, will shed light on new directives aimed at preventing long-term consequences of enteric infection.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.ijpara.2016.11.010.

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